Laurent Perrinet

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Research interests

My research program aims to elucidate the relationship between neural structure and sensory function: how do the topographic organization of neurons and the spiking nature of neural signals enable sensory systems to optimally adapt to the statistical regularities of natural scenes, through predictive processing mechanisms? This work opens the way to novel neuromorphic algorithms with high energy efficiency.

Areas of specialization

Spatio-temporal inference in low-level sensory areas.
Unsupervised learning in topographic maps.
Predictive coding, sparse representations and active perception.

Education

2017 — Habilitation à Diriger des Recherches (HDR), Aix-Marseille Université. Title: "Predictive coding in visuo-motor transformations".

1999–2003 — PhD in Cognitive Science, ONERA/DTIM & Université Paul Sabatier, Toulouse (France). Highest distinction with jury congratulations.

1993–1998 — MSc in Engineering, Supaéro (Toulouse, France). Specialization in stochastic models for signal and image processing.

Professional experience

2020–… — Research Director (DR2, CID51), Institut de Neurosciences de la Timone, CNRS / Aix-Marseille Université, Marseille.

2004–2020 — Research Scientist (CR2 then CR1), INCM then INT, CNRS, Marseille.

2010–2012 — Long-term visit, Karl Friston's theoretical neurobiology group, Wellcome Trust Centre for Neuroimaging, University College London, UK.

2004 — Post-doctoral fellow, Redwood Neuroscience Institute, San Francisco, USA. Supervisor: Bruno Olshausen.

Supervision

6 PhD theses directed (completed), 4 co-directed (completed), 7 current PhD students (5 as principal supervisor, 2 co-supervised), 5 post-docs supervised. Reviewer for 18 PhD defenses.

Selected grants

2023–2027 *Emergences* – Near-physics emerging models for embedded AI (PEPR France 2030).

2021–2025 PI: ANR *AgileNeuRobot* (ANR-20-CE23-0021) – Agile bio-mimetic aerial robots for real-world flight, 435 k€.

2022–2025 PI: *Polychronies* (A*MIDEX AMX-21-RID-025) – Spatio-temporal spike motifs, 100 k€.

2021–2026 Collaborator: ANR *PRIOSENS* (2021–2025), ANR *ACES* (2022–2026), ANR *RubinVase* (2021–2024).

Selected publications

2026 L. U. Perrinet. "Working Memory with Polychronous Chains." ***arXiv preprint arXiv:2604.14096***. doi: 10.48550/arXiv.2604.14096.

2026 J.-N. Jérémie, E. Daucé, L. U. Perrinet. "Foveated Retinotopy Improves Classification and Localization in Convolutional Neural Networks." ***Vision***. doi: 10.3390/vision10020017.

2025 S. Gunasekaran, A. Kembay, H. Ladret, R.-J. Zhu, L. U. Perrinet, O. Kavehei, J. Eshraghian. "A Predictive Approach to Enhance Time-Series Forecasting." ***Nature Communications***. doi: 10.1038/s41467-025-63786-4.

2023 A. Grimaldi, L. U. Perrinet. "Learning heterogeneous delays in a layer of spiking neurons for fast motion detection." ***Biological Cybernetics***. doi: 10.1007/s00422-023-00975-8.

2023 H. Ladret, N. Cortes, L. Ikan, F. Chavane, C. Casanova, L. U. Perrinet. "Cortical recurrence supports resilience to sensory variance in the primary visual cortex." ***Nature Communications Biology***. doi: 10.1038/s42003-023-05042-3.

2021 V. Boutin, A. Franciosini, F. Y. Chavane, F. Ruffier, L. U. Perrinet. "Sparse Deep Predictive Coding captures contour integration capabilities of the early visual system." ***PLoS Computational Biology***. doi: 10.1371/journal.pcbi.1008629.

2020 C. Pasturel, A. Montagnini, L. U. Perrinet. "Humans adapt their anticipatory eye movements to the volatility of visual motion properties." ***PLoS Computational Biology***. doi: 10.1371/journal.pcbi.1007438.

2014 L. U. Perrinet, R. A. Adams, K. Friston. "Active inference, eye movements and oculomotor delays." ***Biological Cybernetics***. doi: 10.1007/s00422-014-0620-8.

2012 K. Friston, R. A. Adams, L. U. Perrinet, M. Breakspear. "Perceptions as Hypotheses: Saccades as Experiments." ***Frontiers in Psychology***. doi: 10.3389/fpsyg.2012.00151.

2012 C. Simoncini, L. U. Perrinet, A. Montagnini, P. Mamassian, G. S. Masson. "More is not always better: dissociation between perception and action explained by adaptive gain control." ***Nature Neuroscience***. doi: 10.1038/nn.3229.

2010 L. U. Perrinet. "Role of homeostasis in learning sparse representations." ***Neural Computation***. doi: 10.1162/neco.2010.05-08-795.

2004 L. U. Perrinet, M. Samuelides, S. J. Thorpe. "Coding static natural images using spiking event times: do neurons cooperate?" ***IEEE Transactions on Neural Networks***. doi: 10.1109/TNN.2004.833303.

Bibliometric indices: 63 peer-reviewed journal articles, Google Scholar: >4400 citations, h=30, i10=64; OpenAlex: 233 documents, >2500 citations, h=26, i10=54. Last updated: October 27, 2026 — Full version.